

MODULE 02

Knee MRI: Grading, Measurements & Instability

Cartilage · Meniscus · Ligaments · TT-TG · Dejour · Ramp Lesions

1.0 AMA PRA Cat 1

Knee

CME Article Review

LEARNING OBJECTIVES

1. Apply Modified Outerbridge grading (0–IV) to articular cartilage on MRI.
2. Use ISAKOS criteria to classify meniscal tears; recognize ramp lesions (Greif).
3. Grade ACL/MCL/PCL injuries and identify secondary signs of rupture.
4. Calculate TT-TG, Insall-Salvati, tibial slope, and trochlear depth.
5. Classify trochlear dysplasia using the Dejour system (A–D).
6. Distinguish major from minor dysplasia and guide surgical planning.

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1. Articular Cartilage — Modified Outerbridge

Grades articular cartilage loss on fluid-sensitive MRI. Report compartment, surface, and size. Grade determines surgical candidacy.

Grade	Depth	Surface	MRI Appearance
0 — Normal	Full thickness	Intact	Homogeneous; smooth
I — Signal	<50%; intact	Intact	Focal T2 blister; surface intact
II — Small	<50%	Partial; <1.5 cm	Partial-thickness fissure
III — Large	>50%; not full	Partial; >1.5 cm	Deep partial; approaching bone
IV — Full	>100%	Full-thickness defect	Down to bone; subchondral edema

KEY POINT

Grade IV in a weight-bearing compartment is an established OA lesion. Grades III–IV are surgical targets for cartilage restoration vs. unloading osteotomy, depending on patient age and alignment.

2. Meniscal Tears — ISAKOS Classification & Ramp Lesions

Tear Pattern & Repair Potential

Pattern	Description	Repair Potential
Vertical longitudinal (peripheral)	Parallel to long axis; red-red zone	Excellent
Radial	Perpendicular; disrupts hoop stress	Poor
Horizontal cleavage	Superior/inferior leaves	Generally poor; repair if young/peripheral
Oblique / Flap	Unstable mobile flap	Partial meniscectomy; repair if large/peripheral
Complex / Degenerative	Multiple planes	Partial meniscectomy; repair rarely feasible

Ramp Lesion — Greif Classification (Posteromedial Meniscocapsular Junction)

Grade	MRI Finding	Stability
0	No abnormality	Stable
1	Partial signal; no discontinuity	Stable
2	Full-thickness signal; no displacement	Potentially unstable

Grade	MRI Finding	Stability
3	Displaced/fluid-distended gap	Unstable — repair required

TEACHING PEARL

Ramp lesions occur in up to 22% of ACL tears. Always inspect the posteromedial corner on coronal and sagittal sequences when ACL tear is identified. The "ghost sign" = absent normal posterior horn at the meniscocapsular junction.

3. Ligament Grading, ACL Signs & Patellofemoral Measurements

MCL Grading (Coronal Fluid-Sensitive MRI)

Grade	Pathology	MRI Findings
I — Sprain	Micro-tears; intact fibers	Periligamentous edema; thickened; continuous fibers
II — Partial	Partial fiber disruption	Partial discontinuity; T2 signal within ligament
III — Complete	Full-thickness rupture	Complete discontinuity; wavy/retracted; fluid gap

Key Patellofemoral Measurements

Measurement	Normal	Threshold	Use
TT-TG distance	<20 mm	>20 mm = medialization osteotomy	Lateral extensor mechanism pull
Insall-Salvati ratio	0.8–1.2	>1.2 = alta; <0.8 = baja	Patellar height
Posterior tibial slope	7–10°	>12° = ACL risk factor	ACL/PCL loading
Trochlear depth	≥3 mm	<3 mm = flat (dysplastic)	Dysplasia screening

Dejour Trochlear Dysplasia (A–D)

Type	Severity	Lateral X-ray Signs	Surgery
A — Minor	Shallow	Crossing sign only	MPFL + TTT
B — Major	Flat/convex	Crossing sign + supratrochlear spur	Trochleoplasty often needed
C — Major	Asymmetric	Crossing sign + double contour	Trochleoplasty
D — Most severe	Asymmetric + cliff	All three signs	Trochleoplasty + MPFL + TTT

References

- Dejour D, Le Coultre B. Sports Med Arthrosc 2007.
- Fardon DF et al. Spine J 2014;14(11):2525–2545.

- Greif DN et al. Knee Surg Sports Traumatol Arthrosc 2020.